1 Integer ambiguity resolution

Integer ambiguity resolution (AR) converts the real-valued, bias-contaminated phase ambiguities of the float solution into the integer carrier-cycle counts they represent. The engine implements MLAMBDA [2] over a between-satellite-differenced ambiguity sub-problem extracted from the filter state and covariance. The operative quantity is the ambiguity differenced against a per-constellation pivot p_g (the differencing section of the Foundations document); the same integer-estimation machinery serves both the single-receiver and the baseline modes, but *what makes that differenced ambiguity an integer differs by mode*. The undifferenced ambiguity $N^i_{r,s}$ of the phase observation equation (Foundations, observation equations) carries a receiver phase bias $b_{r,s}$ and a satellite phase bias b^i_s. In single-receiver PPP / PPP-RTK the pivot difference is a *between-satellite single difference* $\Delta\,\cdot$ (the between-satellite single-difference section of the Foundations document), which cancels the receiver bias $b_{r,s}$ (common to every tracked satellite) but *not* the satellite bias; the residual $\Delta N^{i,p_g}_{r,s}$ is integer only once b^i_s has been removed by a consistent external phase-bias product (BIA / SSR, the SSR-corrections document). In baseline mode the state already holds true double differences $\nabla\Delta\,\cdot$ (the double-difference section of the Foundations document), where the additional between-receiver difference cancels the satellite bias as well, so $\nabla\Delta N$ is integer intrinsically. We retain the resolver's internal label $\nabla\Delta$ for the differenced ambiguity in the equations below — the algebra is identical either way — with the understanding that it denotes the between-satellite single difference Δ in the single-receiver case and the full double difference only in baseline mode. The implementation is split along a driver/kernel seam: `src/core/filter/AmbiguityResolver.cpp` is the *driver* — state-vector assembly, metre↔cycle rescaling, per-pair λ lookup, DD formation, the WL/NL decomposition, and application of the returned integer fix — while the decorrelation-agnostic *numerical core* (the LDL^{T} factorisation, the $\boldsymbol{Z}$-reduction, the Schnorr–Euchner search, the bootstrap success rate, and the full acceptance battery, all as pure free functions over ($\hat{\boldsymbol{N}}$ cycles, $\boldsymbol{Q}_{\hat{N}}$ cycles^2) with no `StateManager` / `KalmanFilter` / config coupling) lives in the `GnssProcessor::ils` namespace of `src/core/filter/IlsCore.{hpp,cpp}`. That same `ils` core is the single implementation shared with the PPP-RTK network producer, which drives the identical battery under its own stricter gate set (**??**). Configuration is under the JSON path `estimation.ambiguity_resolution.*`.

1.1 The integer least-squares problem

The float solution delivers a real-valued ambiguity estimate $\hat{\boldsymbol{N}}$ (in cycles) with covariance $\boldsymbol{Q}_{\hat{N}}$, both obtained by sub-vector extraction from the filter state $\boldsymbol{x}$ and covariance $\boldsymbol{P}$. Integer ambiguity resolution is the weighted closest-lattice-point problem

$$\boxed{\check{\boldsymbol{N}} \;=\; \arg\min_{\boldsymbol{a}\,\in\,\mathbb{Z}^n} \big(\hat{\boldsymbol{N}}-\boldsymbol{a}\big)^{\mathrm{T}}\,\boldsymbol{Q}_{\hat{N}}^{-1}\,\big(\hat{\boldsymbol{N}}-\boldsymbol{a}\big),} \tag{1}$$

i.e. minimisation of the squared Mahalanobis distance from $\hat{\boldsymbol{N}}$ to a lattice point $\boldsymbol{a}$ in the metric of $\boldsymbol{Q}_{\hat{N}}^{-1}$. This is an NP-hard integer least-squares (ILS) program; LAMBDA and its MLAMBDA variant make it tractable in three invariant stages — (i) an LDL^{T} factorisation of $\boldsymbol{Q}_{\hat{N}}$, (ii) a decorrelating integer $\boldsymbol{Z}$-transform (lattice reduction), and (iii) a depth-first shrinking-ellipsoid search over the reduced lattice — none of which carries a tuning parameter. We follow the hat/check convention of the notation block: $\hat{\boldsymbol{N}}$ is the float estimate, $\check{\boldsymbol{N}}$ the integer-fixed solution.

Float extraction and rescaling to cycles. The filter state stores per-band ambiguity states in *metres* ($\lambda_s N$, one per tracked (satellite, signal)), not cycles, so the resolver first forms the between-satellite-differenced float and covariance by explicit differencing against the pivot p_g and then rescales by wavelength. (In baseline mode the state already holds the double difference and the differencing step below collapses to a direct selection; see Section 1.6.) For a differenced pair k with satellite index $i(k)$ and pivot index $p(k)$ (`src/core/filter/AmbiguityResolver.cpp`):

$$y_k^{\nabla\Delta} = \boldsymbol{x}_{\,i(k)} - \boldsymbol{x}_{\,p(k)}, \qquad \text{(DD float, m)} \tag{2}$$

$$\boldsymbol{Q}_{kl}^{\nabla\Delta} = \boldsymbol{P}_{i_k i_l} - \boldsymbol{P}_{i_k p_l} - \boldsymbol{P}_{p_k i_l} + \boldsymbol{P}_{p_k p_l}, \qquad \text{(DD covariance, m}^2\text{)} \tag{3}$$

$$\hat{\boldsymbol{N}}_k = y_k^{\nabla\Delta}/\lambda_k, \qquad (\boldsymbol{Q}_{\hat{N}})_{kl} = \boldsymbol{Q}_{kl}^{\nabla\Delta}/(\lambda_k\lambda_l). \qquad \text{(cycle domain)} \tag{4}$$

The rescaling (4) is load-bearing: running MLAMBDA on metre-domain floats would round toward integer-*metre* values with no carrier-cycle semantics, and the ratio test (Section 1.5) would degenerate to ≈ 1 irrespective of float quality. The per-pair wavelength λ_k is resolved with GLONASS FDMA per-satellite frequencies where applicable, and $\boldsymbol{Q}^{\nabla\Delta}$ is symmetrised, $\boldsymbol{Q}^{\nabla\Delta} \leftarrow \frac{1}{2}(\boldsymbol{Q}^{\nabla\Delta} + \boldsymbol{Q}^{\nabla\Delta\mathrm{T}})$, before factorisation. The DD pivot p_g is the highest-elevation satellite within each (constellation, signal) group, consistent with the BSSD pivot of the differencing section of the Foundations document; all remaining satellites in the group are differenced against it.

1.2 Decorrelation: LDL^{T} factorisation and the integer Z-transform

Factorisation. MLAMBDA factorises the symmetric positive-definite ambiguity covariance as

$$\boldsymbol{Q}_{\hat{N}} \;=\; \boldsymbol{L}^{\mathrm{T}}\,\boldsymbol{D}\,\boldsymbol{L}, \qquad \boldsymbol{L} \text{ unit lower-triangular}, \;\; \boldsymbol{D} = \mathrm{diag}(d_1, \ldots, d_n), \;\; d_i > 0, \tag{5}$$

factored in reverse index order (the Chang–Yang MLAMBDA convention). A non-positive pivot $d_i \le 0$ aborts the fix, since it signals a non-PD covariance. After reduction, d_i is the *conditional variance* of the i-th decorrelated ambiguity given the later ones; it feeds the success-rate gate of Section 1.5 directly.

The Z-transform. The central device of LAMBDA [2] is a unimodular integer transform $\boldsymbol{Z} \in \mathbb{Z}^{n\times n}$, $|\det \boldsymbol{Z}| = 1$, that re-parameterises

$$\boldsymbol{z} = \boldsymbol{Z}^{\mathrm{T}}\hat{\boldsymbol{N}}, \qquad \boldsymbol{Q}_{\hat{z}} = \boldsymbol{Z}^{\mathrm{T}}\,\boldsymbol{Q}_{\hat{N}}\,\boldsymbol{Z}, \tag{6}$$

so that the transformed ambiguities are nearly decorrelated (small off-diagonal $\boldsymbol{L}$ entries) and their conditional variances are ordered. We write the decorrelated float vector as $\bar{\boldsymbol{z}} := \boldsymbol{Z}^{\mathrm{T}}\hat{\boldsymbol{N}}$ (the quantity the search of Section 1.2 actually rounds; implemented as `z_float = Z.transpose()*a` in `ils::mlambda` (`src/core/filter/IlsCore.cpp`)), with components $\bar{z}_k$. Because $\boldsymbol{Z}$ is unimodular the integer minimiser maps back exactly, $\check{\boldsymbol{N}} = (\boldsymbol{Z}^{\mathrm{T}})^{-1}\check{\boldsymbol{z}}$ with $(\boldsymbol{Z}^{\mathrm{T}})^{-1}$ also integer; decorrelation therefore does not alter the solution of (1) — it only makes the search ellipsoid closer to spherical, collapsing the enumeration count by orders of magnitude. The reduction interleaves two operations sweeping $j = n-2$ downward.

Integer Gauss transformations zero the magnitude of sub-diagonal $\boldsymbol{L}$ entries by integer column operations,

$$\mu = \lceil L_{ij} \rfloor, \qquad \boldsymbol{L}_{(:,j)} \mathrel{-}= \mu\,\boldsymbol{L}_{(:,i)}, \qquad \boldsymbol{Z}_{(:,j)} \mathrel{-}= \mu\,\boldsymbol{Z}_{(:,i)}, \tag{7}$$

where $\lceil\cdot\rfloor$ is round-to-nearest, keeping every $|L_{ij}| \le \frac{1}{2}$.

Permutations (conditional adjacent swaps) reorder ambiguities when doing so reduces the later conditional variance. With $\bar{d}_j = d_j + L_{j+1,j}^2\, d_{j+1}$, the swap fires when

$$\bar{d}_j + \varepsilon \;<\; d_{j+1}, \qquad \varepsilon = 10^{-6}, \tag{8}$$

after which d_j, d_{j+1}, the affected rows/columns of $\boldsymbol{L}$ and the columns of $\boldsymbol{Z}$ are updated by the standard rank-one formulas ($\eta = d_j/\bar{d}_j$, $\lambda = d_{j+1}L_{j+1,j}/\bar{d}_j$, $d_j \leftarrow \eta\, d_{j+1}$, $d_{j+1} \leftarrow \bar{d}_j$), and the sweep restarts from $j = n-2$. The guard $\varepsilon = 10^{-6}$ prevents thrashing on near-equal variances and is a numerical safeguard, not an estimator knob; it has no config exposure.

The shrinking-ellipsoid search. The reduced problem is solved by depth-first enumeration of lattice points inside the ellipsoid $\sum_i (z_i - \hat{z}_i)^2/d_i \le \chi^2$, in Schnorr–Euchner order (nearest integer first, then zig-zagging outward). Recursing from the last level $k = n-1$ down to 0,

$$\hat{z}_k = \bar{z}_k + \sum_{i>k} (z_i - \hat{z}_i)\, L_{ik}, \qquad \text{(conditional centre)} \tag{9}$$

$$z_k = \lceil \hat{z}_k \rfloor, \qquad \mathrm{dist}_k = \mathrm{dist}_{k+1} + \frac{(\hat{z}_k - z_k)^2}{d_k}. \qquad \text{(partial distance)} \tag{10}$$

When a complete integer vector reaches level 0 with cumulative distance below the current bound χ^2, it is recorded and the bound shrinks to the worst retained candidate, contracting the ellipsoid as better candidates are found. The search (`ils::search` in `src/core/filter/IlsCore.cpp`) retains the $m = 2$ best candidates (for the ratio test) sorted by ascending squared distance, under a hard iteration cap (`kMaxIter` = 10 000). Its outputs are the two smallest objective values $s_1 \le s_2$ of (1) and the best integer vector $\check{\boldsymbol{z}}$, back-transformed via $\check{\boldsymbol{N}} = (\boldsymbol{Z}^{\mathrm{T}})^{-1}\check{\boldsymbol{z}}$.

1.3 Wide-lane / narrow-lane decomposition (PPP path)

Standard PPP and PPP-RTK AR use a *wide-lane-first* decomposition when Melbourne–Wübbena (MW)-resolved wide-lane (WL) integers are available, halving the search dimension and exploiting the long WL wavelength [1]. The driver attempts the WL-constrained narrow-lane (NL) fix before any unconstrained LAMBDA and — critically — does *not* fall through to an unconstrained search when WL information exists but the constrained fix fails: a wrong-WL fix is worse than no fix, so "no NL AR for satellites whose WL is float" is the enforced hierarchy (`src/core/filter/AmbiguityResolver.cpp`).

WL from Melbourne–Wübbena (integer by contract). The DD wide-lane integer is differenced directly from the MW detector's single-difference WL integers (`src/core/slip/MwSlipDetector`), not searched:

$$\nabla\Delta N_{\mathrm{WL}} = N_{\mathrm{WL}}^{i} - N_{\mathrm{WL}}^{p_g} \in \mathbb{Z}, \qquad \lambda_{\mathrm{WL}} = \frac{c}{f_1 - f_2}. \tag{11}$$

NL search variable. For each dual-frequency DD pair the float narrow-lane ambiguity (ionosphere-free phase combination, in NL cycles) is formed by the transform $\boldsymbol{T}$,

$$\hat{n}_{\mathrm{NL}} = \frac{f_1\,\nabla\Delta N_1 + f_2\,\nabla\Delta N_2}{f_1 + f_2} = \frac{1}{f_1 + f_2}\left(\frac{f_1}{\lambda_1}\,\nabla\Delta_1 + \frac{f_2}{\lambda_2}\,\nabla\Delta_2\right), \qquad \lambda_{\mathrm{NL}} = \frac{c}{f_1 + f_2}, \tag{12}$$

with reduced covariance $\boldsymbol{Q}_{\mathrm{NL}} = \boldsymbol{T}\,\boldsymbol{Q}^{\nabla\Delta}\,\boldsymbol{T}^{\mathrm{T}}$. Only NL-paired DDs enter LAMBDA; unpaired single-signal DDs are excluded so they cannot contaminate the ratio test.

Integer-recentring shift. The float $\hat{n}_{\mathrm{NL}}$ is *not* integer-valued in general: substituting $N_2 = N_1 - \mathrm{WL}$ gives $\hat{n}_{\mathrm{NL}} = N_1 - \frac{f_2}{f_1+f_2}\,\mathrm{WL}$, which for GPS L1/L2 with $\mathrm{WL} = 1$ sits at $N_1 - 0.4379$ — neither integer is meaningfully closer and the ratio test degenerates near 1.6 regardless of float quality. The resolver therefore adds the known-WL offset so the search variable is integer in expectation,

$$\tilde{n}_k = \hat{n}_{\mathrm{NL},k} + \frac{f_2}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL},k} \;\approx\; N_{1,k} \;\in\; \mathbb{Z} + \text{noise}, \tag{13}$$

runs MLAMBDA on $\tilde{\boldsymbol{n}}$ (a constant per-row shift leaves $\boldsymbol{Q}_{\mathrm{NL}}$ unchanged), then back-shifts $\check{n}_{\mathrm{NL}} = \check{N}_1 - \frac{f_2}{f_1+f_2}\,\mathrm{WL}$.

Back-transform to per-signal DDs. The fixed integers map back to the two original signal DDs in metres,

$$\nabla\Delta_1^{\mathrm{fix}} = \Big(\check{n}_{\mathrm{NL}} + \tfrac{f_2}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL}}\Big)\lambda_1, \qquad \nabla\Delta_2^{\mathrm{fix}} = \Big(\check{n}_{\mathrm{NL}} - \tfrac{f_1}{f_1+f_2}\,\nabla\Delta N_{\mathrm{WL}}\Big)\lambda_2, \tag{14}$$

which re-enter the filter through the constrained update of Section 1.6.

State-basis variants. When `estimation.state_ambiguity_basis = nl_wl` the state holds NL/WL ambiguities directly and there are no per-band DD pairs to enumerate; today only the MW-resolved WL integers are applied as constraint rows (a full NL LAMBDA search with WL pre-constrained is future work in this basis). The complementary WL-only mode (`estimation.ambiguity_resolution.wl_only`, in the `per_band` basis) applies the cycle-domain constraint $\nabla\Delta_1/\lambda_1 - \nabla\Delta_2/\lambda_2 = \nabla\Delta N_{\mathrm{WL}}$ as a single pseudo-observation per pair without any NL search, with covariance inflation on the affected single-difference states when MW re-locks at a new integer so the state snaps within one epoch. Both WL-constraint paths (this one and the `nl_wl`-basis constraint below) build their pseudo-observation noise from a *separate* knob, $\boldsymbol{R} = \sigma_{\mathrm{wl}}^2\,\boldsymbol{I}$ with $\sigma_{\mathrm{wl}}^2 =$ `wl_only_pseudo_var` (10^{-4} cycles^2, note the cycle domain), *not* the metre-domain σ_{hold}^2 of the NL/DD feedback in (19). This `per_band wl_only` path carries no cross-frequency consistency check: it gates each DD WL only on the per-arc fractional residual (`wl_max_frac`, default 0.40, both arcs). The cross-frequency parity check lives instead in the `nl_wl` basis (`applyMwWlConstraintsNlWl_` in `src/core/filter/AmbiguityResolver.cpp`), where it rejects all three WL DDs of a triple-frequency satellite when the telescoping canonical cycle-count consistency $\nabla\Delta_{ab} - \nabla\Delta_{ac} + \nabla\Delta_{bc} \neq 0$ fails; it does *not* run in the `per_band wl_only` path.

1.4 Partial / subset ambiguity resolution

Partial AR operates at *two* levels: a static upstream candidate gate that screens which ambiguities ever enter the integer problem, and an optional dynamic backward-elimination search that recovers a fixable subset when the full set fails validation.

Upstream candidate gating (always on). Subset selection begins when the DD pairs are built (`src/core/filter/AmbiguityResolver.cpp`). A satellite/signal ambiguity is admitted to the integer set only if it passes every filter:

- **Initialised** — the state entry is non-zero (seeded by data), and **non-frozen** — its variance is non-zero (excludes a BSSD-frozen pivot).
- **Elevation** — $\theta_r^i \geq \theta_{\mathrm{AR}}$ (`elevation_mask`).

- **Resolvable** — the satellite is not flagged unresolvable for want of bias/ATX (BIA, ATX) coverage (single-receiver path; in baseline DD the analogous filter is the `exclude_keys` set of non-integer-capable DDs from `MeasurementModel`).
- **Constellation** — $g(i) \in$ `active_systems`, when that list is non-empty.
- **Pairable** — the (constellation, signal) group holds ≥ 2 satellites, so a pivot difference exists.

The constellation filter keeps a system in the *float* solution while excluding it from the candidate set — useful when one constellation's residual model bias would drag down the ratio test.

Dynamic backward-elimination subset (`enable_partial_ar`, default off). With the candidate set fixed, the default behaviour is all-or-nothing: the full set is searched and either passes all of Section 1.5 or the epoch stays float. When `enable_partial_ar` is set, a failure of the full-set battery triggers a greedy backward elimination (`ils::selectFixableSubset` in `src/core/filter/IlsCore.cpp`, reached through the driver-side `selectFixableSubset_` config adapter) instead of falling straight back to float. Starting from the full candidate set, each iteration re-runs the complete MLAMBDA pipeline of Sections 1.1 to 1.2 on the current subset and re-tests the *same* acceptance battery — ratio lower/upper bound (15), bootstrap success rate (16), and per-entry fractional cap (17). On any failure it drops one entry and retries, down to a single ambiguity:

- if the *fractional cap* is the failing gate, the entry with the largest $|\hat{\boldsymbol{N}}_k - \check{\boldsymbol{N}}_k|$ (nearest a half-cycle boundary) is removed;
- otherwise the entry with the largest formal float variance $(\boldsymbol{Q}_{\hat{N}})_{kk}$ — the least precise ambiguity, whose removal most improves both the bootstrap rate and the search conditioning — is removed.

Because the accepted subset is always re-validated by the full battery on its own covariance sub-block, this can only fix *fewer* ambiguities than the full set, never a worse-conditioned set: a wrong elimination order costs fixes, not integrity, and the path is bit-identical to the legacy behaviour when off. The same routine serves the per-band SD, baseline DD, and WL-constrained NL paths uniformly, operating purely in the decorrelation-agnostic cycle domain; each caller maps the surviving indices back to its own state rows and feeds them through the constrained update of Section 1.6.

1.5 Validation: ratio test and fixed-failure-rate gate

A fix is committed only if it passes all three of the following gates (a logical AND). The ratio and fixed-failure-rate (FFR) gates are theoretically distinct: one bounds the discrimination between the two best candidates, the other bounds the absolute success probability.

Ratio test (discrimination).

$$\boxed{\text{ratio} = \frac{s_2}{s_1} \;\geq\; \tau_{\text{ratio}}, \qquad s_1 \leq s_2 \text{ the two smallest ILS objectives,}} \tag{15}$$

with s_1, s_2 from the search of Section 1.2 and threshold τ_{ratio} (`ratio_threshold`). A second upper-bound rejection ratio $> \tau_{\text{ratio}}^{\max}$ (`ratio_upper_bound`) catches the pathological case of a degenerate covariance direction — a near-zero s_1 left by a previous wrong-integer hold — that would otherwise produce ultra-confident but meaningless ratios.

Fixed-failure-rate gate (bootstrap success rate). A geometry-driven, self-tuning lower bound on the ILS success probability,

$$\boxed{P_{s,\mathrm{boot}} = \prod_{i=1}^{n} \mathrm{erf}\left(\frac{1}{2\sqrt{2\,d_i}}\right) \;\geq\; P_s^{\min},} \tag{16}$$

where the d_i are the post-reduction LDL^{T} conditional variances of (5) and $P_s^{\min}$ is `min_success_rate`. Per dimension the factor is $\Pr\big(|\mathcal{N}(0, d_i)| < \frac{1}{2}\big)$ — the probability that sequential bootstrapped rounding lands on the correct integer — and the product is the standard Teunissen lower bound on the true ILS success rate. This ties the acceptance threshold to the actual problem geometry rather than to a fixed ratio alone.

Per-pair residual backstop. The joint ratio test can pass on a borderline set in which one pair sits near a half-cycle boundary; the fix is rejected if any single DD's fractional residual exceeds a cap,

$$\max_k \big|\hat{\boldsymbol{N}}_k - \check{\boldsymbol{N}}_k\big| \;\leq\; \kappa_{\mathrm{frac}} \quad \text{(cycles)}, \tag{17}$$

with $\kappa_{\mathrm{frac}} =$ `nl_max_per_pair_frac`.

Convergence guards. Before any search the resolver short-circuits AR while the float is immature — if the epoch count is below `min_epochs` or the position standard deviation $\sqrt{\boldsymbol{P}_{00} + \boldsymbol{P}_{11} + \boldsymbol{P}_{22}}$ exceeds `max_position_sigma_m` — because a marginal fix accepted before the covariance settles can lock onto wrong integers and then poison every subsequent epoch.

1.6 Feedback: constrained pseudo-observation update

Accepted integers are *not* fed back by the classical perfect back-substitution (the $R = 0$ conditioning form)

$$\boldsymbol{x}_{\mathrm{fix}} = \boldsymbol{x} - \boldsymbol{Q}_{\boldsymbol{aa}}\, \boldsymbol{Q}^{\nabla\Delta-1}\big(y^{\nabla\Delta} - \check{\boldsymbol{N}}\big), \qquad \boldsymbol{P}_{\mathrm{fix}} = \boldsymbol{P} - \boldsymbol{Q}_{\boldsymbol{aa}}\, \boldsymbol{Q}^{\nabla\Delta-1}\boldsymbol{Q}_{\boldsymbol{aa}}^{\mathrm{T}}, \tag{18}$$

but via a *finite-variance Joseph-form pseudo-observation* (`src/core/filter/AmbiguityResolver.cpp`; kernel `KalmanFilter::josephPseudoUpdate` in `include/filter/KalmanFilter.hpp`). The reason is structural: (18) drives the fixed ambiguity rows of $\boldsymbol{P}$ to ≈ 0 variance, so the next epoch's LAMBDA sees a near-singular $\boldsymbol{Q}^{\nabla\Delta}$ and emits meaningless inflated ratios that trip the upper-bound guard of (15) indefinitely. A finite-variance pseudo-observation instead keeps $\boldsymbol{P}$ well-conditioned and lets subsequent epochs re-confirm the integer (asymptoting toward it) or detect a slip through its residual.

For each accepted DD pair k the constraint row, innovation and noise are

$$\boxed{\boldsymbol{H}_{(k,\,i)} = +1, \quad \boldsymbol{H}_{(k,\,p)} = -1, \qquad v_k = \lambda_k\, \check{\boldsymbol{N}}_k - \big(\boldsymbol{x}_i - \boldsymbol{x}_p\big), \qquad \boldsymbol{R} = \sigma_{\mathrm{hold}}^2\, \boldsymbol{I},} \tag{19}$$

followed by $(\boldsymbol{x}_{\mathrm{fix}}, \boldsymbol{P}_{\mathrm{fix}}) = \textsc{JosephUpdate}(\boldsymbol{x}, \boldsymbol{P}, \boldsymbol{H}, \boldsymbol{v}, \boldsymbol{R})$, with $\sigma_{\mathrm{hold}}^2 =$ `hold_variance`. The DD/baseline path uses a single-column selector $\boldsymbol{H}_{(k,\,\mathrm{idx})} = +1$ since the corresponding state value already *is* the double difference. Fix-and-hold mode (`estimation.ambiguity_resolution.mode` $= 3$) re-applies the same constraint each epoch to suppress float drift. This pseudo-observation feedback is the resolution of the mode-3 destabilisation that the earlier $R = 0$ form (18) induced; (18) survives in the code only as the documented, rejected legacy approach.

1.7 Configuration summary

All tunable thresholds live under the JSON path `estimation.ambiguity_resolution.*`, parsed in `src/core/config/GnssConfigLoad.cpp` with defaults in `include/filter/AmbiguityResolver.hpp`. The invariant estimator theory of Sections 1.1 to 1.6 carries no knobs — changing it changes the mathematics, not a policy.

Symbol	Config key (`...ambiguity_resolution.`)	Default	Role
—	`mode`	`0`	0 float / 1 continuous / 2 instantaneous / 3 fix-and-hold
τ_{ratio}	`ratio_threshold`	`3.0`	ratio-test acceptance (15)
$\tau_{\text{ratio}}^{\max}$	`ratio_upper_bound`	`1e4`	degenerate-covariance reject (15)
$P_s^{\min}$	`min_success_rate`	`0.999`	FFR gate (16)
κ_{frac}	`nl_max_per_pair_frac`	`0.15`	per-pair residual cap (17)
θ_{AR}	`elevation_mask`	`20°`	candidate elevation mask
σ^2_{hold}	`hold_variance`	`0.001` m^2	pseudo-obs $\boldsymbol{R}$ (19)
$n^{\text{ep}}_{\min}$	`min_epochs`	`10`	convergence guard
$\sigma^{\max}_{\text{pos}}$	`max_position_sigma_m`	`0.05` m	convergence guard
—	`wl_only`	`false`	MW-WL-only, no NL search
σ^2_{wl}	`wl_only_pseudo_var`	`1e-4` cyc^2	WL-constraint pseudo-obs $\boldsymbol{R}$ (Section 1.3)
—	`wl_max_frac`	`0.40`	per-arc WL fractional-residual gate (Section 1.3)
—	`enable_partial_ar`	`false`	backward-elimination subset fix (Section 1.4)
—	`active_systems`	`[]` (all)	constellation subset (Section 1.4)
—	`state_ambiguity_basis` †	`per_band`	`per_band` vs `nl_wl` dispatch

† Path is `estimation.state_ambiguity_basis` — one level above the `ambiguity_resolution` block.

References

[1] D. Laurichesse et al. "Integer ambiguity resolution on undifferenced GPS phase measurements and its application to PPP and satellite precise orbit determination". In: *Navigation* 56.2 (2009), pp. 135–149. DOI: 10.1002/j.2161-4296.2009.tb01750.x.

[2] P. J. G. Teunissen. "The least-squares ambiguity decorrelation adjustment: a method for fast GPS integer ambiguity estimation". In: *Journal of Geodesy* 70.1–2 (1995), pp. 65–82. DOI: 10.1007/BF00863419.